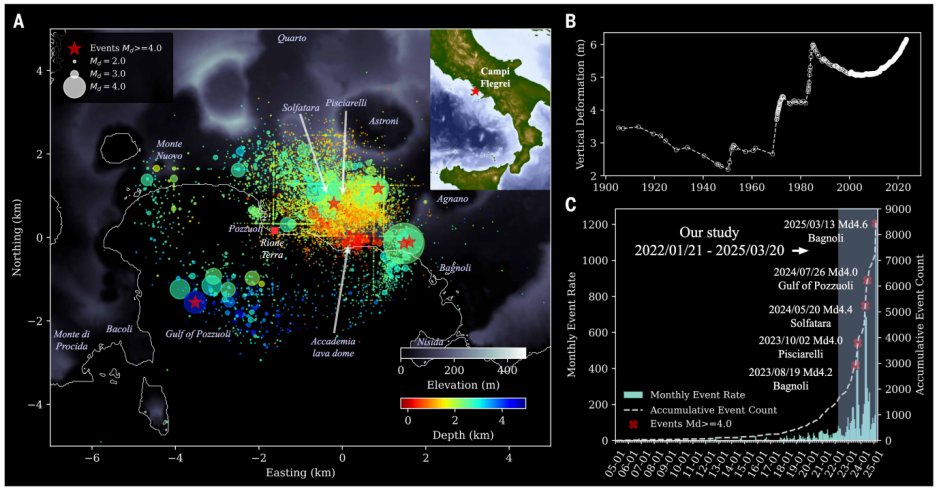
Figure 1: Seismicity and ground displacement

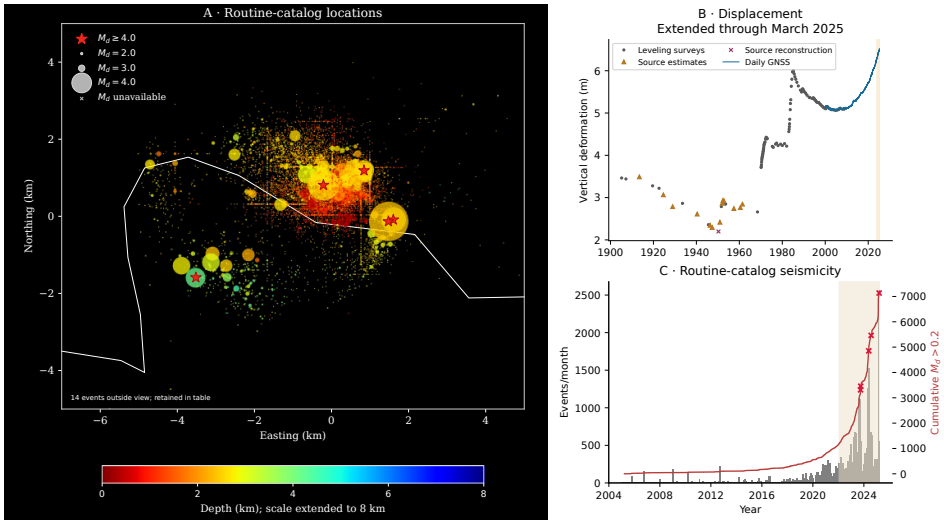
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

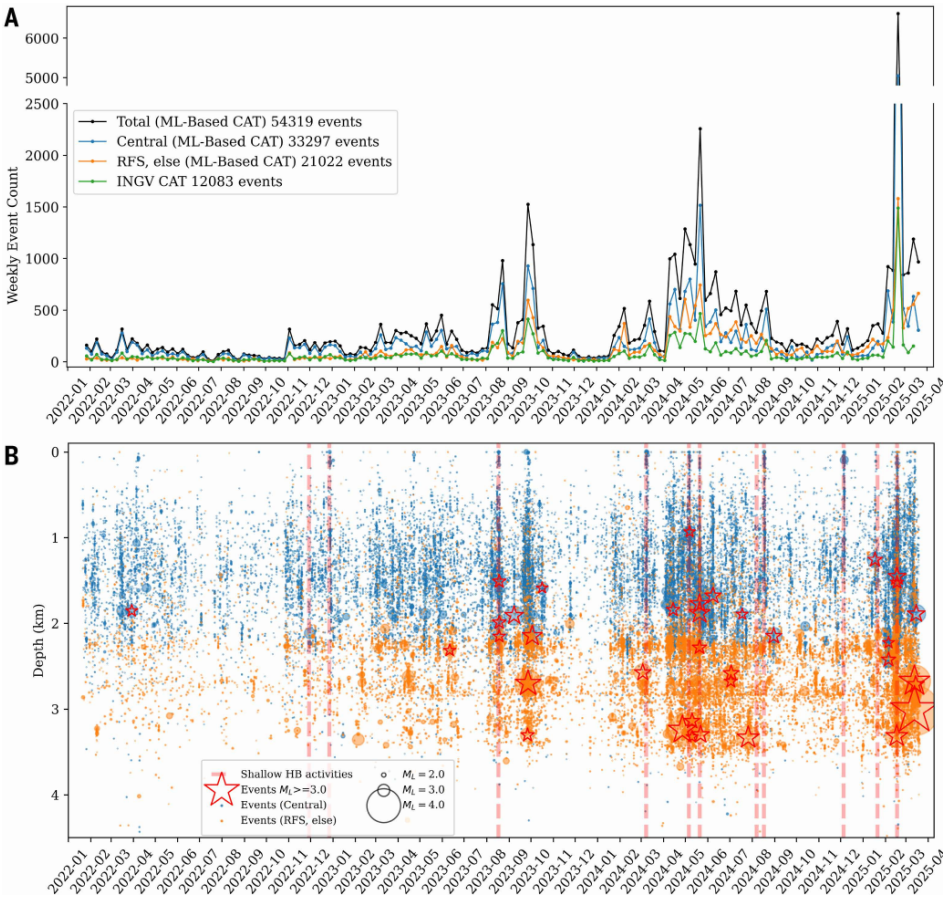
Panel 1A–C — partially reproduced



Partial contextual reconstruction. A: external routine catalog, 16,046 located events and five Md>=4 stars. B: 114 leveling surveys (circles), 13 source-provided estimates (triangles), one source-provided reconstruction (cross), and daily GNSS through 20 March 2025; extension beyond 2023 shaded. Historical estimates are not raw observations or reconstructions made here. C: all routine monthly counts and cumulative Md>0.2; March includes days 1–20 only. Routine events are not independent detections. Datum joining is an assumption; basemap and live snapshot differ from the publication. Depth colors run red to blue on a dark background; the color scale extends to 8 km to include the observed range. Circle area includes an exponential Md term and a small visibility offset, an explicit display assumption; crosses indicate missing Md. Published pixels are not plotting inputs.

Figure 2: Temporal evolution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 3.
DOI: 10.1126/science.adw9038.

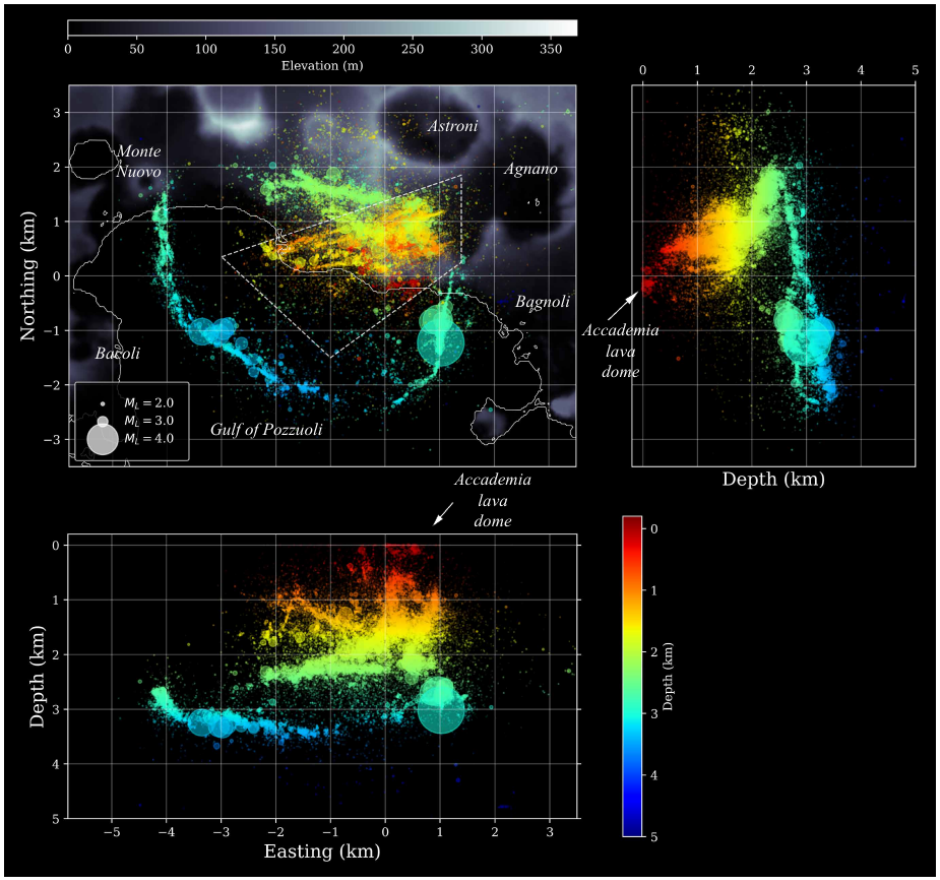
Independent result

Missing: 2A, 2B — not reproduced

Panels A and B need multiyear calibrated/relocated events and magnitudes. The frozen 1003-event calibration cohort and the expanded associations remain provisional.

Figure 3: Spatial distribution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

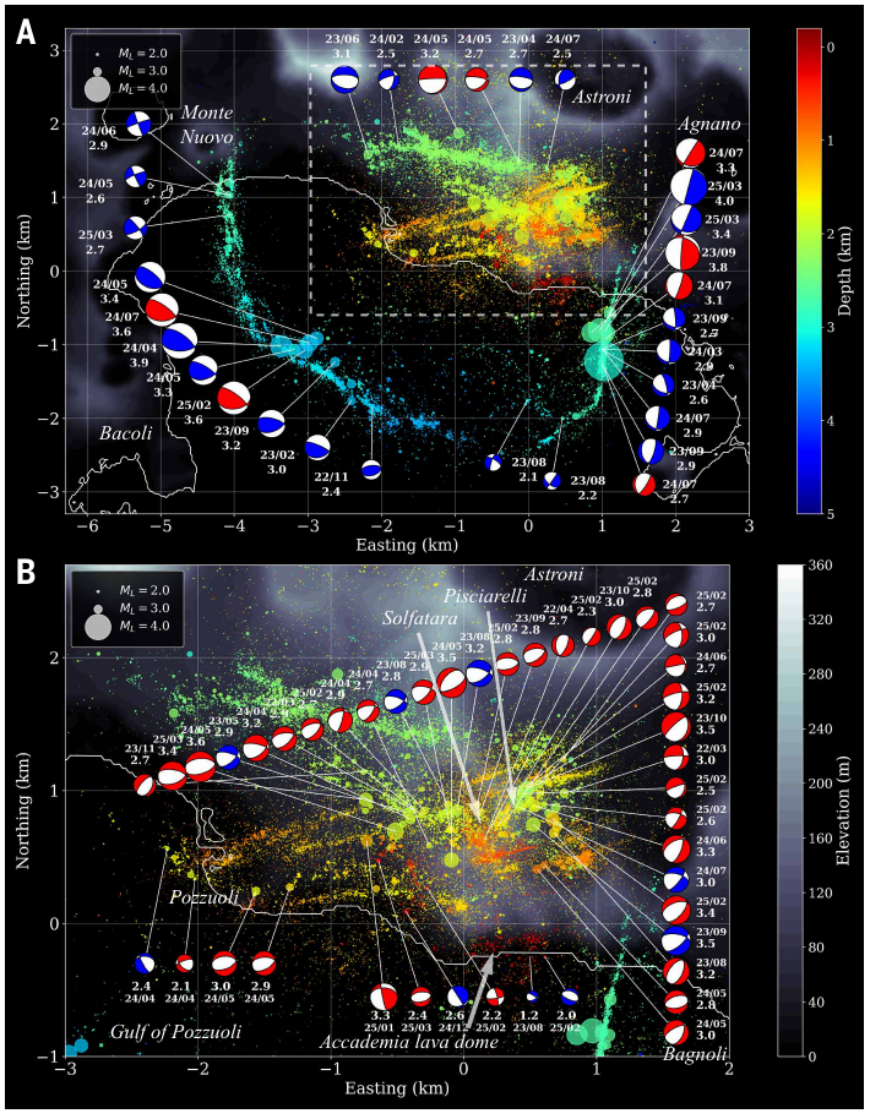
Independent result

Missing: 3/map, 3/northing-depth, 3/easting-depth — not reproduced

The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 4: Focal mechanisms of larger events

Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

Missing: 4A, 4B — not reproduced

Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Figure S1: Recorded station availability

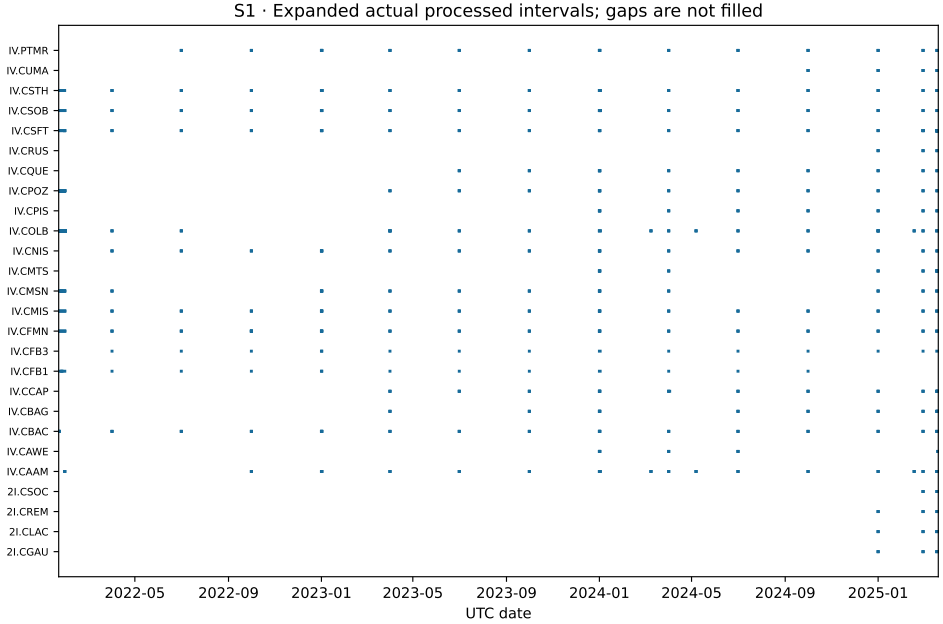
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

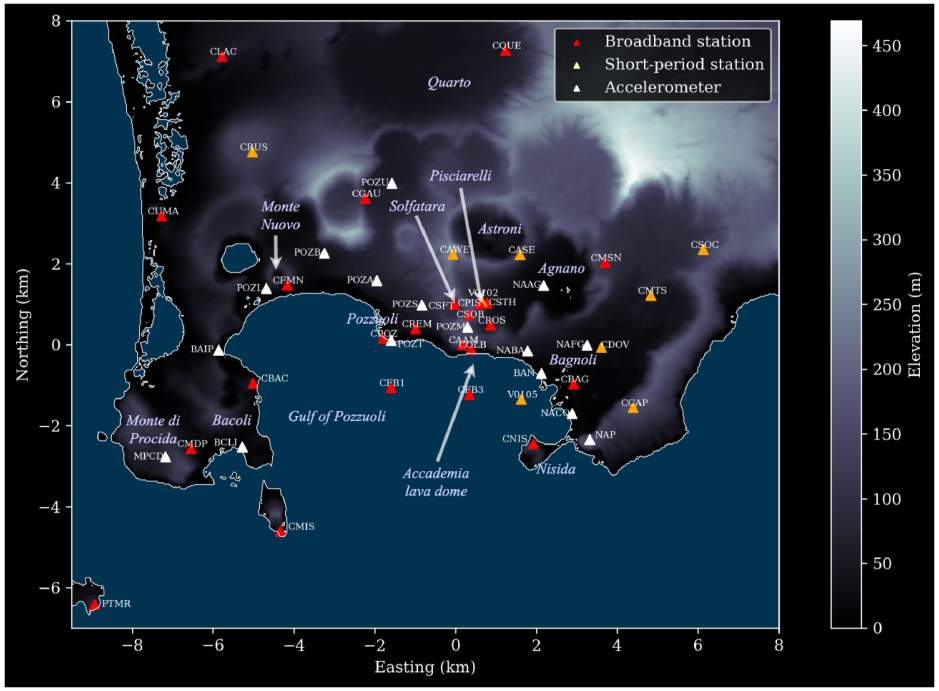
Panel S1 — partially reproduced



Partial: expanded checkpoint, 868 processed files, 26 station names, 8511.520 union channel-hours. Empty intervals are neither connected nor counted. The separately frozen calibration cohort retains its older event identifiers.

Figure S2: Station locations

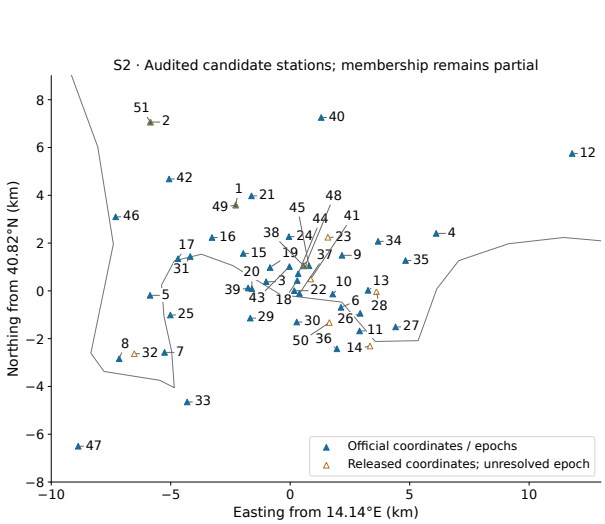
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

Panel S2 — partially reproduced



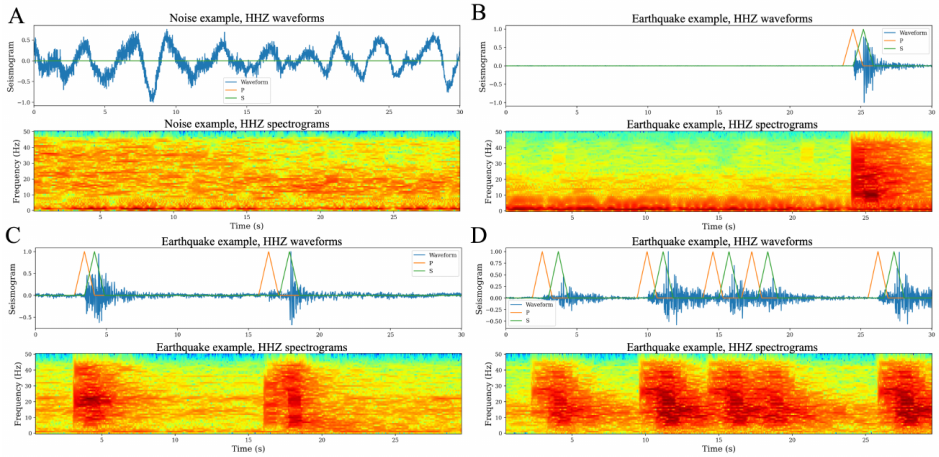
1	2I.CGAU..HH	27	IV.CCAP..EH
2	2I.CLAC..HH	28	IV.CDOV..EH
3	2I.CREM..HH	29	IV.CFB1..HH
4	2I.CSOC..EH	30	IV.CFB3..HH
5	IT.BAIP..HN	31	IV.CFMN..HH
6	IT.BAN..HN	32	IV.CMDP..HH
7	IT.BCLI..HN	33	IV.CMIS..HH
8	IT.MPCD..HN	34	IV.CMSN..HH
9	IT.NAAG..HN	35	IV.CMTS..EH
10	IT.NABA..HN	36	IV.CNIS..HH
11	IT.NACO..HN	37	IV.COLB..HH
12	IT.NAD..HN	38	IV.CPIS..HH
13	IT.NAFG..HN	39	IV.CPOZ..HH
14	IT.NAP..HN	40	IV.CQUE..HH
15	IT.POZA..HN	41	IV.CROS..HH
16	IT.POZB..HN	42	IV.CRUS..EH
17	IT.POZL..HN	43	IV.CSFT..HH
18	IT.POZM..HN	44	IV.CSOB..HH
19	IT.POZS..HN	45	IV.CSTH..HH
20	IT.POZT..HN	46	IV.CUMA..HH
21	IT.POZU..HN	47	IV.PTMR..HH
22	IV.CAAM..HH	48	IV.V0102..EH
23	IV.CASE..EH	49	IV.V0104..HH
24	IV.CAWE..EH	50	IV.V0105..EH
25	IV.CBAC..HH	51	IV.V0106..HH
26	IV.CBAG..HH		

42 official epoch matches; 9 unresolved.
51 identifiers ≠ verified 48 instruments.
Two coordinate-coincident pairs are retained.
See station_geometry_qa.csv for every entry.

Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

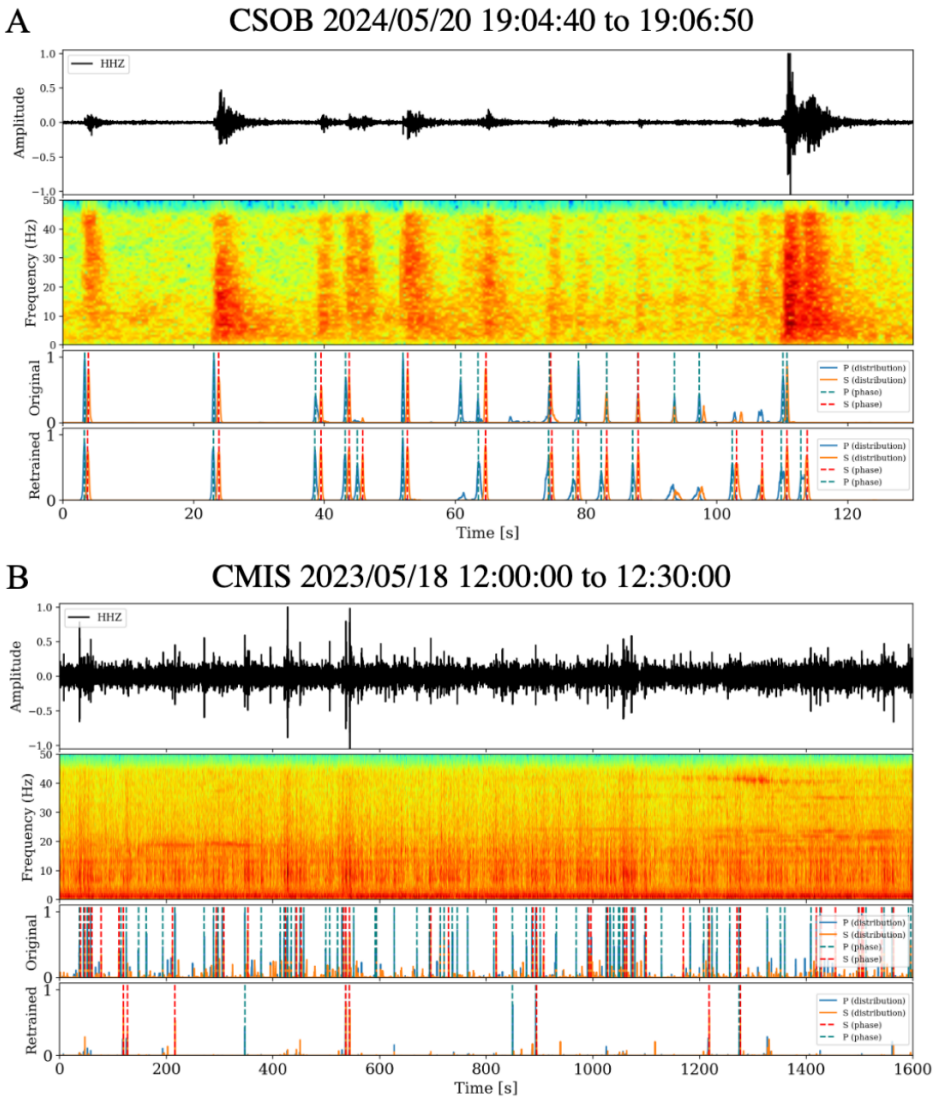
Independent result

Missing: S3A, S3B, S3C, S3D — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

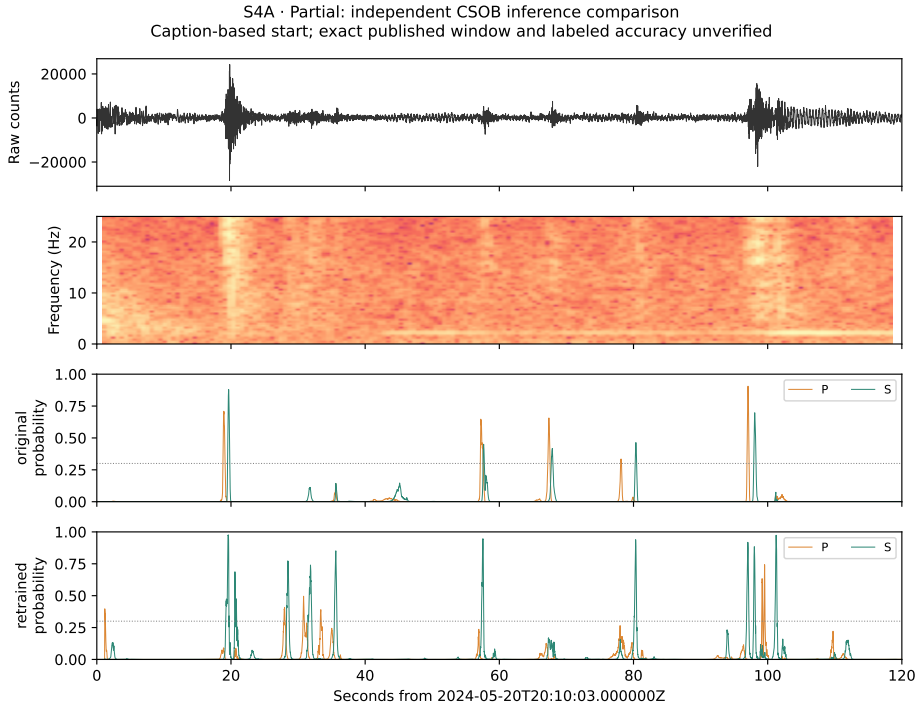
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

Independent result

Panel S4A — partially reproduced



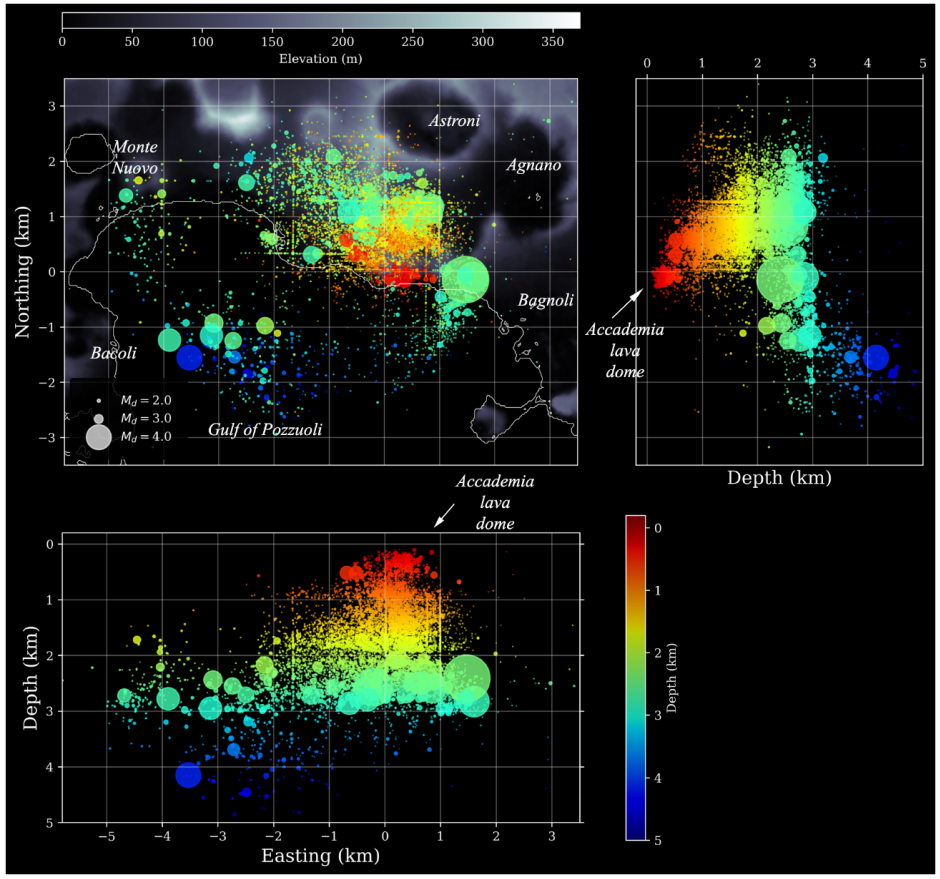
Partial S4A inference comparison on original CSOB observations, 20 May 2024 20:10:03–20:12:03 UTC. The caption's two-hour offset is interpreted literally; exact published alignment is unverified. Original weights: 5 P and 5 S; retrained weights: 6 P and 10 S. No labeled accuracy or retraining claim.

Missing: S4B — not reproduced

S4B: exact CMIS noise-window timing and reference labels are unavailable. Other calendar-window diagnostics are not inserted as that panel.

Figure S5: Routine-catalog locations

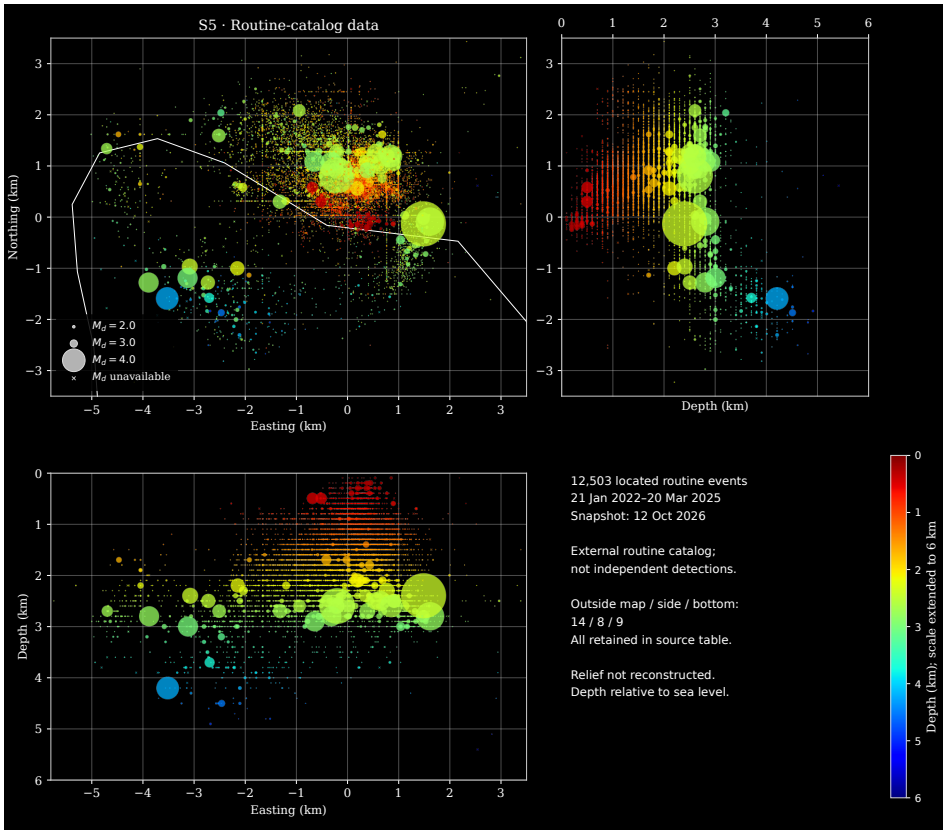
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

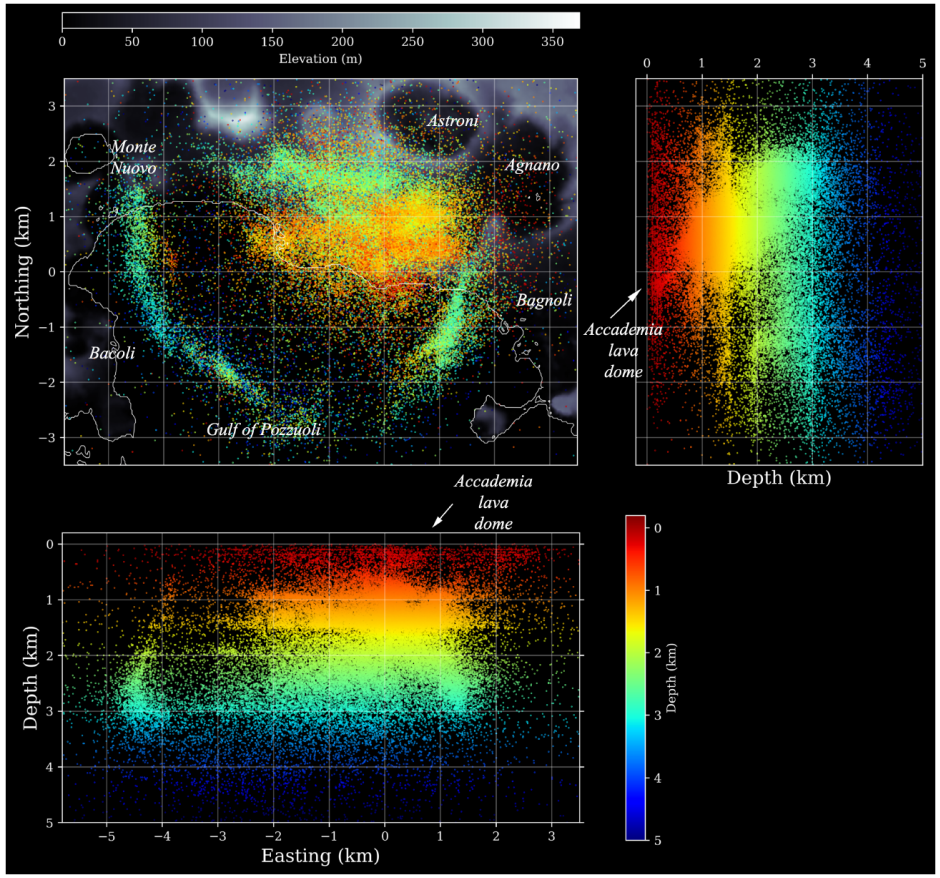
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections. Depth colors run red to blue on a dark background; the color scale extends to 6 km to include the observed range. Circle area includes an exponential Md term and a small visibility offset, an explicit display assumption; crosses indicate missing Md. Published pixels are not plotting inputs. The public source reports depths in 0.1 km steps; the resulting bands are not jittered or smoothed.

Figure S6: Initial VELEST locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

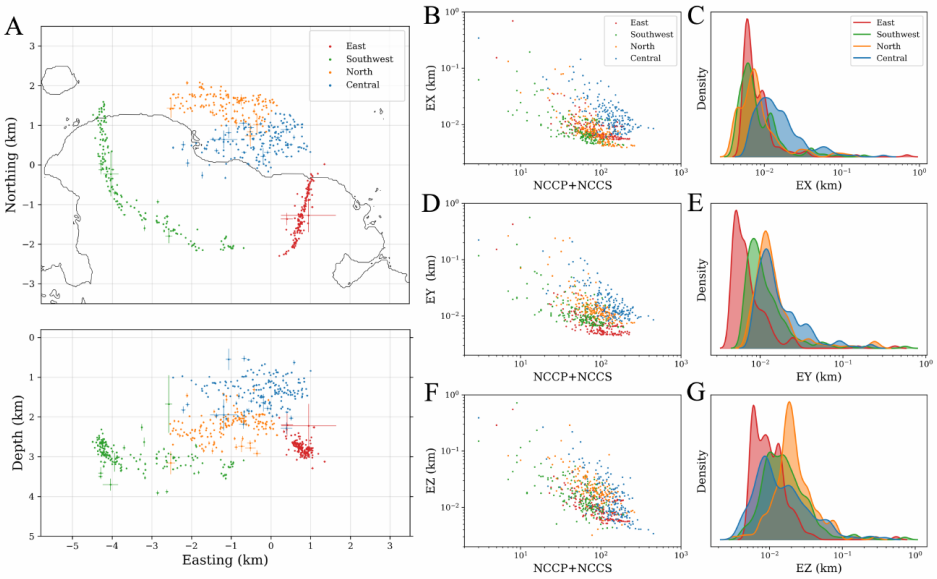
Independent result

Missing: S6/map, S6/northing-depth, S6/easting-depth — not reproduced

No validated independent counterpart. Alternative-profile VELEST and HypoDD trials are retained as diagnostics after the supplement: calibration fails convergence/physical checks and one SVD subset diverges. The fixed 3D background is unverified; TomoDD and the specified regional uncertainty analysis are not completed.

Figure S7: Location uncertainties

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

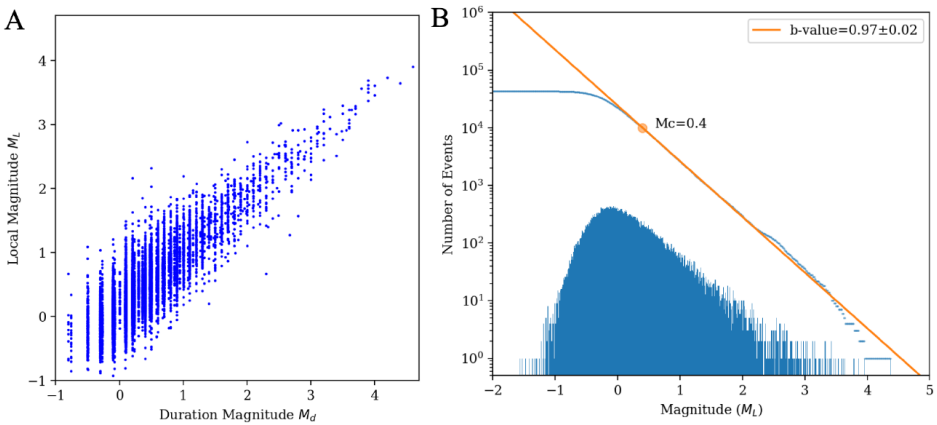
Independent result

Missing: S7A/map, S7A/easting-depth, S7B, S7D, S7F, S7C, S7E, S7G — not reproduced

No validated independent counterpart. Alternative-profile VELEST and HypoDD trials are retained as diagnostics after the supplement: calibration fails convergence/physical checks and one SVD subset diverges. The fixed 3D background is unverified; TomoDD and the specified regional uncertainty analysis are not completed.

Figure S8: Magnitude calibration and frequency distribution

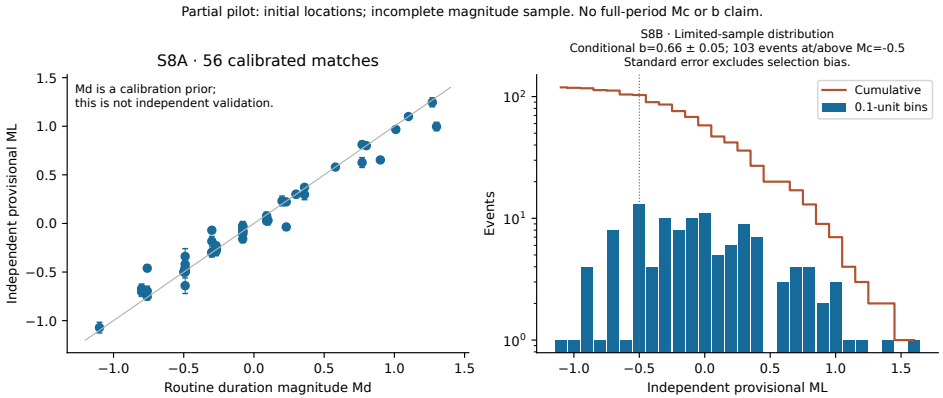
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

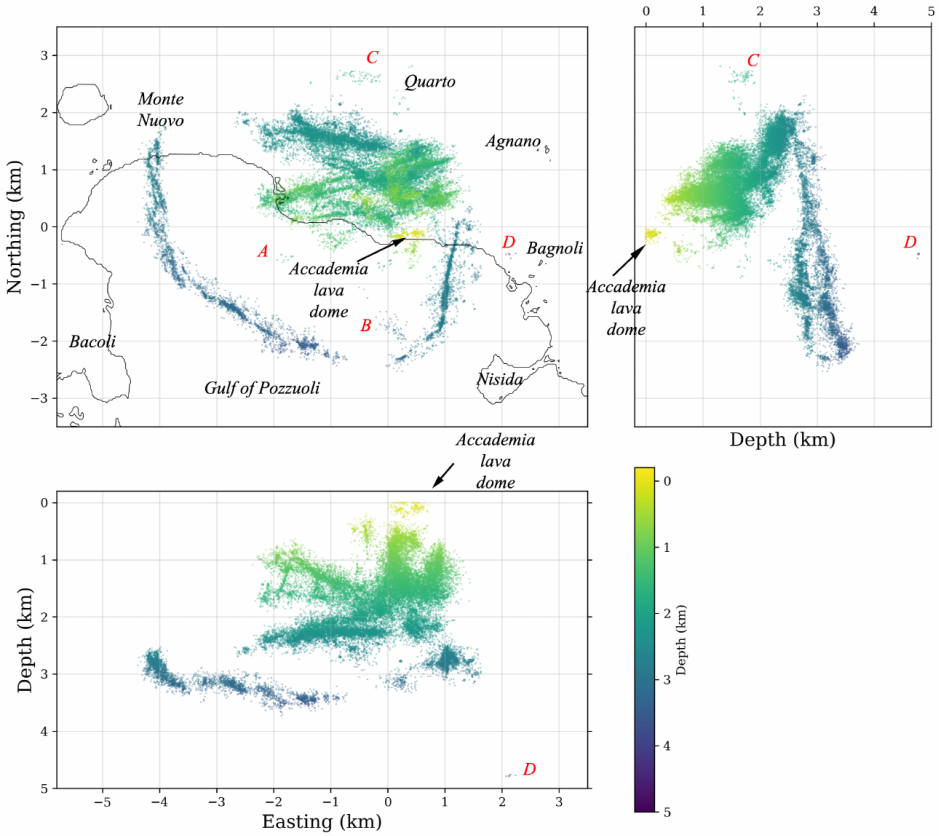
Panel S8A–B — partially reproduced



Partial amplitude-ratio pilot: 119 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

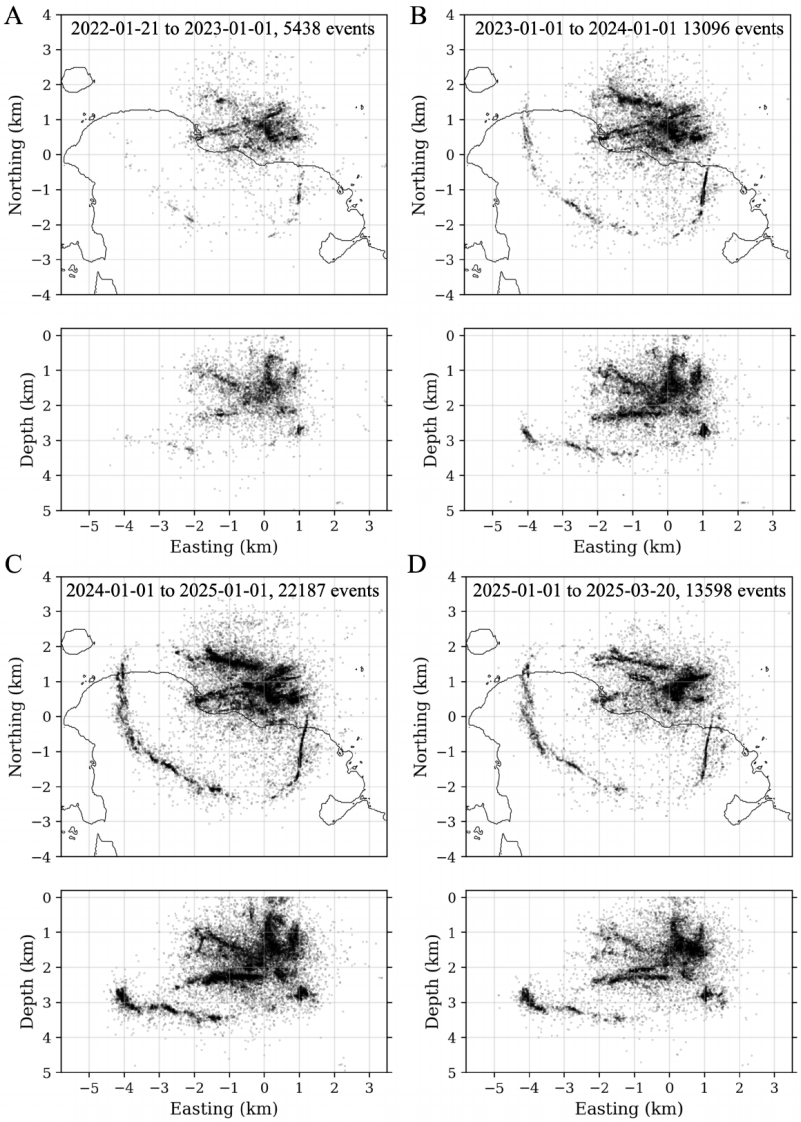
Independent result

Missing: S9/map, S9/northing-depth, S9/easting-depth — not reproduced

All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

Figure S10: Seismicity by year

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

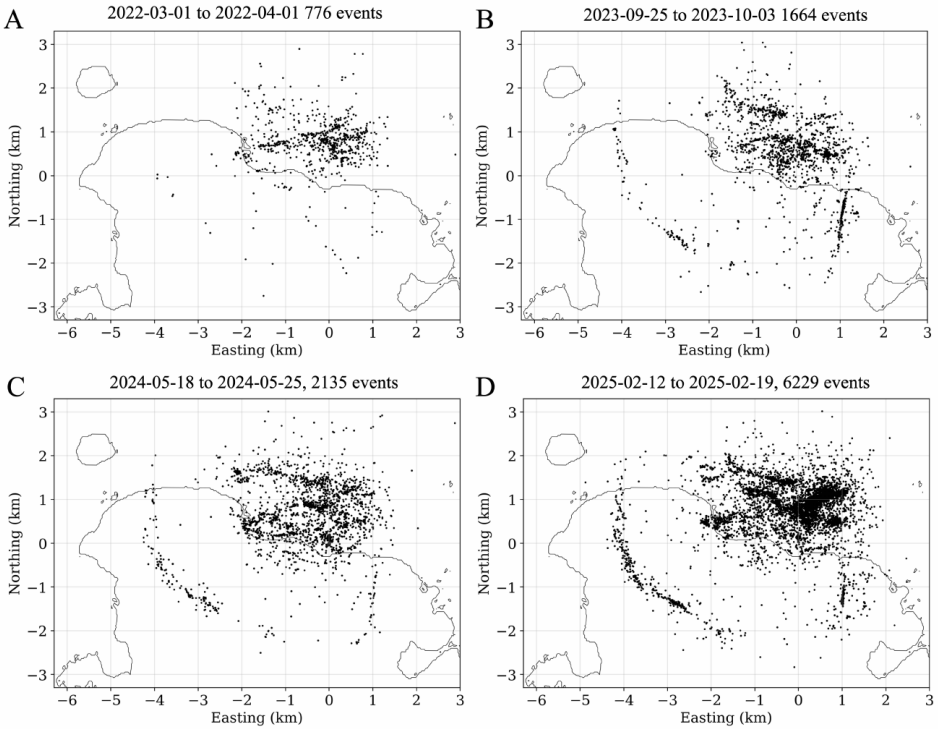
Independent result

Missing: S10/2022, S10/2023, S10/2024, S10/2025 — not reproduced

The four annual groups for 2022–2025 require the independent multi-year relocated catalog. The available sample does not support them.

Figure S11: Four major swarms

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

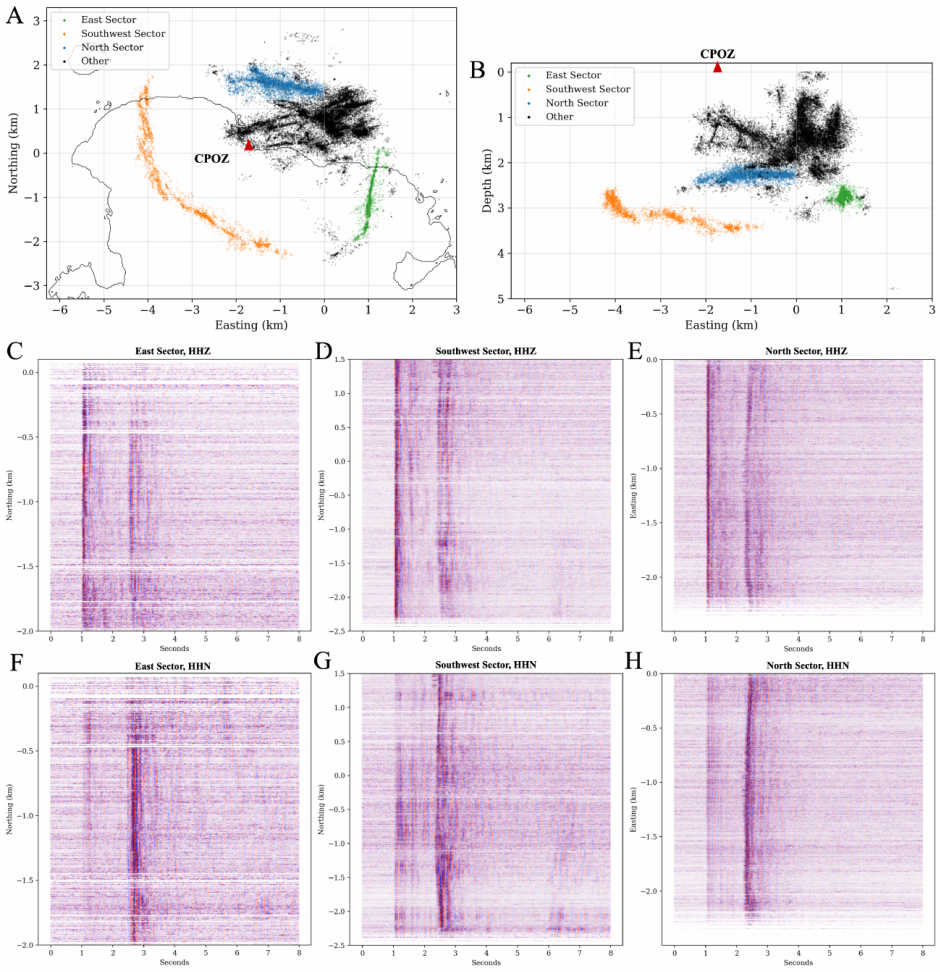
Independent result

Missing: S11/swarm-1, S11/swarm-2, S11/swarm-3, S11/swarm-4 — not reproduced

The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S12: Ring sectors and CPOZ common-receiver gathers

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

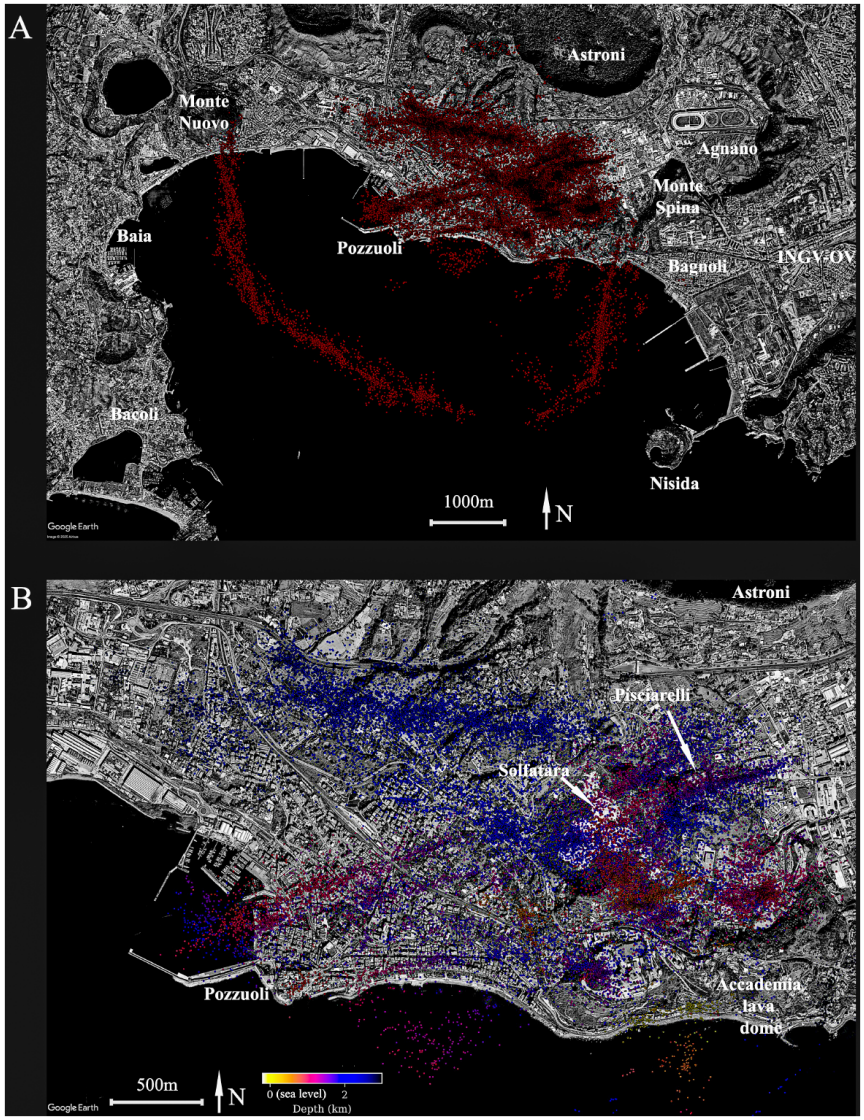
Independent result

Missing: S12A, S12B, S12C, S12D, S12E, S12F, S12G, S12H — not reproduced

Panels A–H need relocated sector selections and manual depth review. 137 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

Figure S13: Relocated seismicity on satellite imagery

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

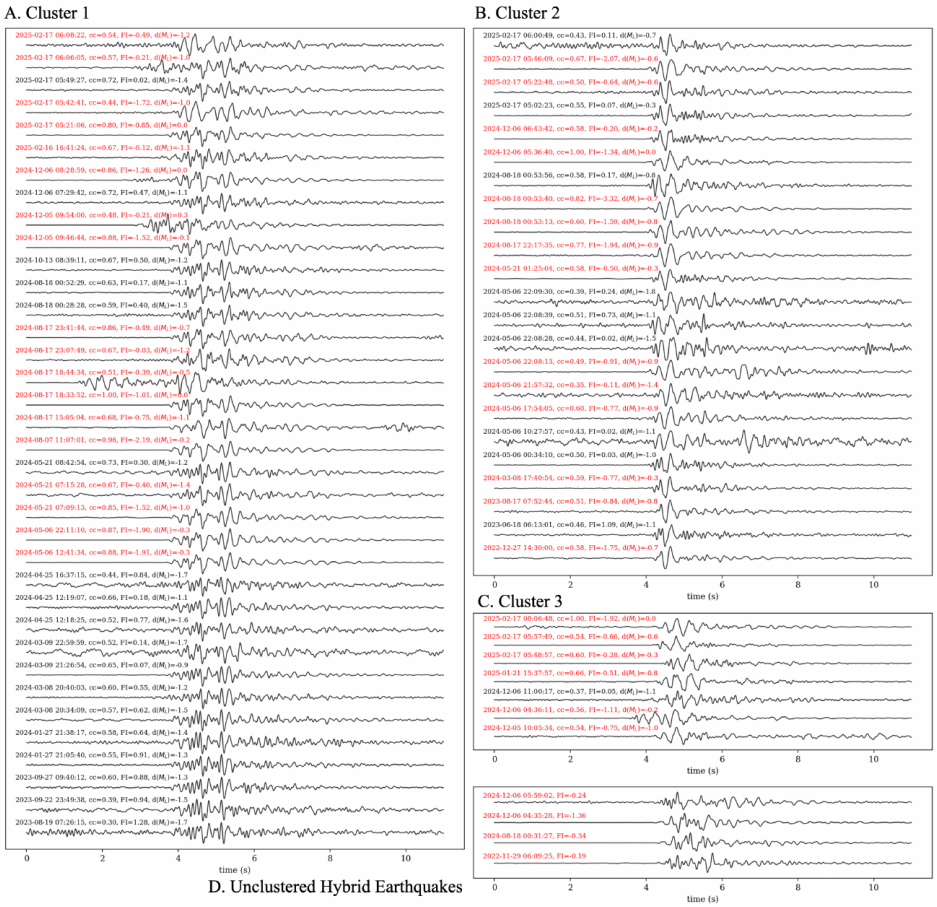
Independent result

Missing: S13A, S13B — not reproduced

Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

Figure S14: Hybrid-event clusters

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

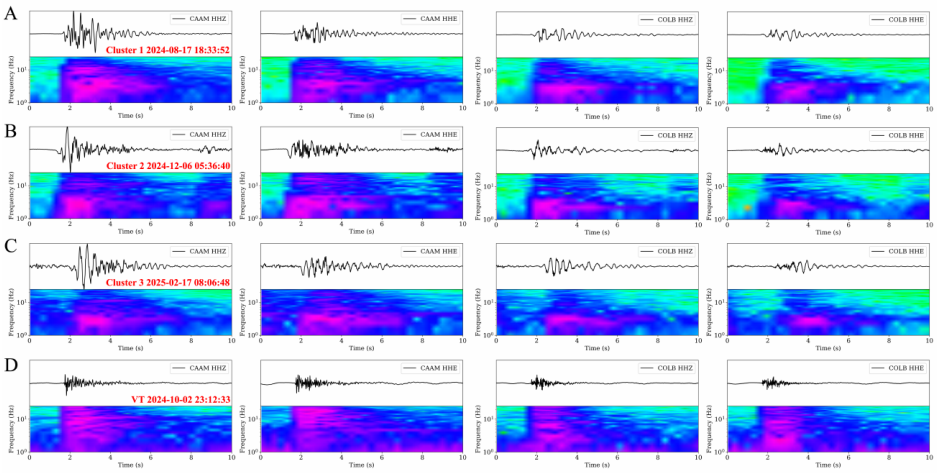
Independent result

Missing: S14/1, S14/2, S14/3, S14/unclustered — not reproduced

The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 75.797 hour scan produced 822 unconfirmed threshold crossings, not reviewed families.

Figure S15: Hybrid and shallow-event waveforms

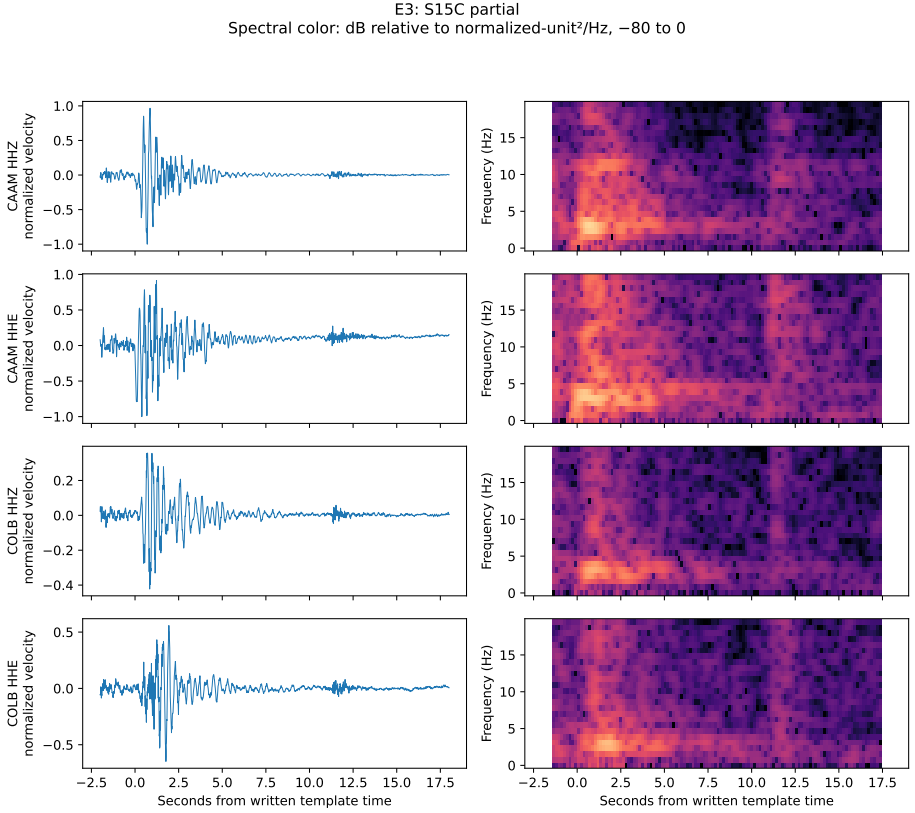
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

Panel S15C — partially reproduced



Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Missing: S15A, S15B, S15D — not reproduced

Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.